

Solution Requirements

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (what the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No login is required in the current prototype; each student uses the app as an open, single-session Streamlit tool on their own device.	Low
2	Authorization levels	No differentiated user roles exist yet; every user has the same access to submit text and view their own analytics dashboard.	Low
3	External interfaces	Integrates with the Google Gemini API to generate natural, personalized guidance text based on the detected emotion and confidence score.	High
4	Transactions processing	Each student submission is processed as a single request: text input, emotion prediction, guidance generation, and CSV log write, completed within one Streamlit interaction cycle.	High
5	Reporting	Generates a live analytics dashboard with a bar chart of emotion-state distribution and a box plot comparing BiLSTM vs. BERT confidence, rendered via Plotly.	Medium
6	Business rules, etc.	Only the five defined emotion categories (Confused, Frustrated, Curious, Confident, Bored) drive response selection; any unmatched case falls back to a generic encouragement message.	High
7	Compliance to laws or regulations	No student personal identifiers are collected; only free-text input and predicted emotion/confidence are logged, limiting personal-data exposure.	Medium
8	Other	Supports side-by-side comparison of BiLSTM and BERT predictions for the same input via a <code>compare_models()</code> function.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Emotion prediction and guidance generation should feel conversational within a single Streamlit interaction.	Response returned in under 3 seconds per submission.

2	Scalability	The system should support a full classroom of students using the tool concurrently without slowdown.	Supports at least 50 concurrent Streamlit sessions in a classroom pilot.
3	Security & Data Privacy	Student text should not be exposed beyond the Gemini API call itself, and the API key must never enter source control.	GEMINI_API_KEY stored only in .env, excluded via .gitignore.
4	Reliability & Availability	If the Gemini API is unavailable, the platform should still return a helpful response rather than failing outright.	100% of requests return a response, falling back to the mock handler on API failure.
5	Usability & Accessibility	The interface should be usable by a student with no training, with clearly labeled emotion and guidance sections.	A first-time user can submit input and read guidance within 30 seconds, with no instructions needed.
6	Other	Maintainability: the four modules (emotion detection, AI response, logging, analytics) should remain independently testable and swappable (e.g., mock vs. real Gemini handler).	Each module can be unit-tested in isolation using mock dependencies.